Synopsis

Project Title => Design and develop an end-to-end machine learning classification system to predict sentiment polarity from product review text data. The project should involve selecting a suitable dataset of product reviews and performing preprocessing steps such as removing noise (HTML tags, punctuation), tokenization, stop word removal, stemming/lemmatization, and converting text into numerical form.

Problem Id: 16

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- **Problem Description:**

In this project, a machine learning-based sentiment classification system is developed to analyze product review text data. The system processes the raw review text, cleans and converts it into a suitable format, and then uses machine learning algorithms to predict the sentiment of the review.

This approach helps companies save time, analyze large amounts of customer feedback efficiently, and make better business decisions based on customer opinions. It is widely used in e-commerce platforms, product marketing, and customer feedback analysis.

- **Steps of Implementation:**

STEPS OF IMPLEMENTATION

Building an ML Pipeline for Sentiment Analysis from Product Reviews

1.DATA COLLECTION=>

Gather a dataset of product reviews with sentiment labels (positive, negative, neutral)

2.DATA PREPROCESSING=>

Clean the text data by removing noise, tokenizing, removing stop words and applying stemming/lemmatization

3.TEXT VECTORIZATION=>

Convert the processed text into numerical vectors using TH-IDF or Bag of Words techniques

4.MODEL SELECTION and TRAINING=>

Choose and train machine learning classifiers like Logistic Regression, KNN (K-Nearest Neighbours), and SVM (Support Vector Machine)

5.MODEL EVALUATION=>

Evaluate t6he trained models using accuracy, precision, recall, F1-score ,confusion matrix, and ROC

Output=> POSITIVE | NEUTRAL | NEGATIVE

Fig. 1 Sample architecture

- **Expected Modules/ Libraries to be used :**
  1. NumPy:- A numerical computing library used for handling arrays and performing mathematical operations efficiently.

  2. Pandas:- A data manipulation library used to read and process datasets (CSV files). It provides DataFrame structures for easy data cleaning, filtering, and analysis.

  3. Scikit-learn:- A machine learning library used to build and evaluate models. It provides algorithms like Logistic Regression, KNN, and SVM, as well as tools for feature extraction and model evaluation.

  4. Matplotlib:- A visualization library used to create graphs and performance charts for model evaluation.

  5. Seaborn:- A statistical data visualization library built on Matplotlib. It is mainly used to plot heatmaps and confusion matrices in an easy and visually clear way.

- **Expected Evaluation metrics and Plots to be used:**

Evaluation Metrics :-

- Accuracy: Overall correct predictions
- Precision: Correct positive predictions
- Recall: Finds all actual positives
- F1-score: Balance of precision & recall
- Confusion Matrix: Shows correct & wrong predictions
- ROC Curve: Shows model performance

Plots :-

- Confusion Matrix (heatmap)
- ROC Curve
- Bar graph (Accuracy comparison of models)
- Bar graph (Precision, Recall, F1-score)

- **Required topics from the subject:**

1.Data Preprocessing:- Data preprocessing is the process of cleaning and preparing raw data before applying machine learning algorithms.

2. Data Standardization:- Standardization is a technique used to scale features so that they have a mean of 0 and a standard deviation of 1.

3.Logistic Regression:- Logistic Regression is a supervised learning classification algorithm used to predict categorical outcomes.

4. K-Nearest Neighbors (KNN)

KNN is a classification algorithm that assigns a class label based on the majority class of its nearest neighbors in the dataset.

- **Platform Required:**

Visual studio/ jupyter notebook/ python idle.

- **Books and Link Sources:**

Machine Learning by Sridhar

Artificial intelligence by Trivedi, Munesh

https://share.google/pYfzUfzWECUwgEDDN

https://share.google/GesQSvzacyaa4Mlzv

https://share.google/QJDh2LL1Yefn8sZ8M

https://share.google/y0YC9W9TgB3CA8k1h